

Insurance Cost Analysis

```
insurance <- read.csv("data/medical_insurance.csv")  
names(insurance)
```

[1] "person_id"	"age"
[3] "sex"	"region"
[5] "urban_rural"	"income"
[7] "education"	"marital_status"
[9] "employment_status"	"household_size"
[11] "dependents"	"bmi"
[13] "smoker"	"alcohol_freq"
[15] "visits_last_year"	"hospitalizations_last_3yrs"
[17] "days_hospitalized_last_3yrs"	"medication_count"
[19] "systolic_bp"	"diastolic_bp"
[21] "ldl"	"hba1c"
[23] "plan_type"	"network_tier"
[25] "deductible"	"copay"
[27] "policy_term_years"	"policy_changes_last_2yrs"
[29] "provider_quality"	"risk_score"
[31] "annual_medical_cost"	"annual_premium"
[33] "monthly_premium"	"claims_count"
[35] "avg_claim_amount"	"total_claims_paid"
[37] "chronic_count"	"hypertension"
[39] "diabetes"	"asthma"
[41] "copd"	"cardiovascular_disease"
[43] "cancer_history"	"kidney_disease"
[45] "liver_disease"	"arthritis"
[47] "mental_health"	"proc_imaging_count"
[49] "proc_surgery_count"	"proc_physio_count"
[51] "proc_consult_count"	"proc_lab_count"
[53] "is_high_risk"	"had_major_procedure"

What proportion of insurance cost variance is attributable to modifiable lifestyle choices versus baseline demographic and socioeconomic risk factors?

```
#Ran a regression for modifiable life choices - might need to switch to glm if assumptions are met
m1 <- lm(annual_premium ~ bmi + smoker + alcohol_freq, data=insurance)
summary(m1)
```

Call:

```
lm(formula = annual_premium ~ bmi + smoker + alcohol_freq, data = insurance)
```

Residuals:

Min	1Q	Median	3Q	Max
-536.0	-220.4	-112.2	83.1	10204.1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	655.3464	9.3629	69.994	<2e-16 ***
bmi	3.4341	0.2497	13.754	<2e-16 ***
smokerFormer	-141.0053	4.6245	-30.491	<2e-16 ***
smokerNever	-192.4150	3.8799	-49.593	<2e-16 ***
alcohol_freqNone	-5.2296	6.0193	-0.869	0.385
alcohol_freqOccasional	-5.8825	5.8747	-1.001	0.317
alcohol_freqWeekly	-8.8204	6.2373	-1.414	0.157

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 394.3 on 99993 degrees of freedom

Multiple R-squared: 0.02619, Adjusted R-squared: 0.02614

F-statistic: 448.3 on 6 and 99993 DF, p-value: < 2.2e-16

```
# 1.DATA ANALYSIS
# View data structure
glimpse(insurance)
```

Rows: 100,000

Columns: 54

```
$ person_id      <int> 75722, 80185, 19865, 76700, 92992, 76435, ~
$ age            <int> 52, 79, 68, 15, 53, 63, 36, 21, 53, 28, 34~
$ sex            <chr> "Female", "Female", "Male", "Male", "Male"~
$ region         <chr> "North", "North", "North", "North", "Centr~
```

\$ urban_rural	<chr> "Suburban", "Urban", "Rural", "Suburban", ~
\$ income	<dbl> 22700, 12800, 40700, 15600, 89600, 305000, ~
\$ education	<chr> "Doctorate", "No HS", "HS", "Some College"~
\$ marital_status	<chr> "Married", "Married", "Married", "Married"~
\$ employment_status	<chr> "Retired", "Employed", "Retired", "Self-em~
\$ household_size	<int> 3, 3, 5, 5, 2, 3, 1, 3, 1, 4, 2, 2, 3, 4, ~
\$ dependents	<int> 1, 1, 3, 3, 0, 2, 0, 2, 0, 2, 1, 0, 1, 2, ~
\$ bmi	<dbl> 27.4, 26.6, 31.5, 31.6, 30.5, 20.4, 21.6, ~
\$ smoker	<chr> "Never", "Never", "Never", "Never", "Never"~
\$ alcohol_freq	<chr> "None", "Weekly", "None", "None", "Daily",~
\$ visits_last_year	<int> 2, 2, 1, 0, 3, 1, 0, 3, 0, 0, 8, 2, 2, 1, ~
\$ hospitalizations_last_3yrs	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ days_hospitalized_last_3yrs	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ medication_count	<int> 4, 3, 4, 1, 2, 2, 0, 1, 0, 0, 3, 3, 0, 2, ~
\$ systolic_bp	<dbl> 121, 131, 160, 104, 136, 137, 114, 82, 138~
\$ diastolic_bp	<dbl> 76, 79, 84, 68, 83, 96, 76, 64, 84, 62, 65~
\$ ldl	<dbl> 123.8, 97.3, 129.5, 160.3, 171.0, 69.2, 15~
\$ hba1c	<dbl> 5.28, 4.82, 5.51, 8.50, 5.20, 5.70, 5.54, ~
\$ plan_type	<chr> "PPO", "POS", "HMO", "HMO", "POS", "HMO", ~
\$ network_tier	<chr> "Bronze", "Gold", "Platinum", "Silver", "P~
\$ deductible	<int> 1000, 1000, 500, 500, 500, 500, 500, 2000,~
\$ copay	<int> 20, 10, 20, 20, 10, 20, 10, 20, 10, 10, 20~
\$ policy_term_years	<int> 4, 1, 10, 5, 7, 5, 3, 6, 4, 4, 10, 9, 5, 2~
\$ policy_changes_last_2yrs	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ provider_quality	<dbl> 3.73, 3.10, 3.90, 3.89, 3.90, 4.66, 4.30, ~
\$ risk_score	<dbl> 0.5714, 1.0000, 1.0000, 0.2857, 0.8681, 0.~
\$ annual_medical_cost	<dbl> 6938.06, 1632.61, 7661.01, 5130.27, 1700.7~
\$ annual_premium	<dbl> 876.05, 445.10, 1538.02, 820.63, 500.93, 5~
\$ monthly_premium	<dbl> 73.00, 37.09, 128.17, 68.39, 41.74, 44.06,~
\$ claims_count	<int> 1, 4, 0, 0, 1, 1, 0, 2, 0, 0, 11, 2, 1, 3,~
\$ avg_claim_amount	<dbl> 4672.59, 297.27, 0.00, 0.00, 1002.24, 1220~
\$ total_claims_paid	<dbl> 4672.59, 1189.08, 0.00, 0.00, 1002.24, 122~
\$ chronic_count	<int> 1, 2, 3, 1, 2, 1, 0, 1, 1, 0, 2, 1, 0, 0, ~
\$ hypertension	<int> 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, ~
\$ diabetes	<int> 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
\$ asthma	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ copd	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ cardiovascular_disease	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, ~
\$ cancer_history	<int> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ kidney_disease	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ liver_disease	<int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
\$ arthritis	<int> 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, ~
\$ mental_health	<int> 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, ~

```

$ proc_imaging_count      <int> 1, 0, 1, 1, 2, 0, 1, 2, 0, 0, 0, 1, 0, 1, ~
$ proc_surgery_count      <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
$ proc_physio_count       <int> 2, 1, 2, 0, 1, 0, 1, 0, 1, 2, 1, 1, 1, 0, ~
$ proc_consult_count      <int> 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, ~
$ proc_lab_count          <int> 1, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 2, 0, ~
$ is_high_risk            <int> 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, ~
$ had_major_procedure     <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~

```

```
summary(insurance)
```

```

      person_id      age      sex      region
Min.   :      1  Min.   : 0.00  Length:100000  Length:100000
1st Qu.: 25001  1st Qu.: 37.00  Class :character  Class :character
Median : 50000  Median : 48.00  Mode  :character  Mode  :character
Mean   : 50000  Mean   : 47.52
3rd Qu.: 75000  3rd Qu.: 58.00
Max.   :100000  Max.   :100.00

      urban_rural      income      education      marital_status
Length:100000  Min.   : 1100  Length:100000  Length:100000
Class :character  1st Qu.: 21100  Class :character  Class :character
Mode  :character  Median : 36200  Mode  :character  Mode  :character
              Mean   : 49874
              3rd Qu.: 62200
              Max.   :1061800

      employment_status  household_size  dependents      bmi
Length:100000  Min.   :1.000  Min.   :0.0000  Min.   :12.00
Class :character  1st Qu.:2.000  1st Qu.:0.0000  1st Qu.:23.60
Mode  :character  Median :2.000  Median :1.0000  Median :27.00
              Mean   :2.431  Mean   :0.8984  Mean   :26.99
              3rd Qu.:3.000  3rd Qu.:1.0000  3rd Qu.:30.40
              Max.   :9.000  Max.   :7.0000  Max.   :50.40

      smoker      alcohol_freq      visits_last_year
Length:100000  Length:100000  Min.   : 0.000
Class :character  Class :character  1st Qu.: 1.000
Mode  :character  Mode  :character  Median : 2.000
              Mean   : 1.928
              3rd Qu.: 3.000
              Max.   :25.000

      hospitalizations_last_3yrs  days_hospitalized_last_3yrs  medication_count
Min.   :0.00000  Min.   : 0.0000  Min.   : 0.000
1st Qu.:0.00000  1st Qu.: 0.0000  1st Qu.: 0.000
Median :0.00000  Median : 0.0000  Median : 1.000

```

Mean :0.09364	Mean : 0.3734	Mean : 1.236	
3rd Qu.:0.00000	3rd Qu.: 0.0000	3rd Qu.: 2.000	
Max. :3.00000	Max. :21.0000	Max. :11.000	
systolic_bp	diastolic_bp	ldl	hba1c
Min. : 61.0	Min. : 40.0	Min. : 30.0	Min. : 3.540
1st Qu.:107.0	1st Qu.: 67.0	1st Qu.: 99.4	1st Qu.: 5.160
Median :117.0	Median : 73.0	Median :120.0	Median : 5.440
Mean :117.8	Mean : 73.6	Mean :120.0	Mean : 5.606
3rd Qu.:128.0	3rd Qu.: 79.0	3rd Qu.:140.3	3rd Qu.: 5.760
Max. :183.0	Max. :114.0	Max. :248.3	Max. :11.940
plan_type	network_tier	deductible	copay
Length:100000	Length:100000	Min. : 500	Min. :10.00
Class :character	Class :character	1st Qu.: 500	1st Qu.:10.00
Mode :character	Mode :character	Median :1000	Median :20.00
		Mean :1227	Mean :19.52
		3rd Qu.:2000	3rd Qu.:30.00
		Max. :5000	Max. :50.00
policy_term_years	policy_changes_last_2yrs	provider_quality	risk_score
Min. : 1.000	Min. :0.00000	Min. :1.500	Min. :0.0000
1st Qu.: 3.000	1st Qu.:0.00000	1st Qu.:3.200	1st Qu.:0.3297
Median : 6.000	Median :0.00000	Median :3.600	Median :0.5055
Mean : 5.518	Mean :0.05069	Mean :3.599	Mean :0.5198
3rd Qu.: 8.000	3rd Qu.:0.00000	3rd Qu.:4.010	3rd Qu.:0.7033
Max. :10.000	Max. :2.00000	Max. :5.000	Max. :1.0000
annual_medical_cost	annual_premium	monthly_premium	claims_count
Min. : 55.55	Min. : 211.7	Min. : 17.64	Min. : 0.000
1st Qu.: 1175.12	1st Qu.: 352.1	1st Qu.: 29.34	1st Qu.: 0.000
Median : 2082.57	Median : 463.6	Median : 38.63	Median : 1.000
Mean : 3009.45	Mean : 582.3	Mean : 48.53	Mean : 1.622
3rd Qu.: 3707.96	3rd Qu.: 666.7	3rd Qu.: 55.56	3rd Qu.: 2.000
Max. :65724.90	Max. :10962.5	Max. :913.55	Max. :23.000
avg_claim_amount	total_claims_paid	chronic_count	hypertension
Min. : 0.0	Min. : 0.0	Min. :0.0000	Min. :0.0000
1st Qu.: 0.0	1st Qu.: 0.0	1st Qu.:0.0000	1st Qu.:0.0000
Median : 318.0	Median : 642.5	Median :1.0000	Median :0.0000
Mean : 656.5	Mean : 1377.9	Mean :0.7247	Mean :0.2034
3rd Qu.: 872.2	3rd Qu.: 1795.5	3rd Qu.:1.0000	3rd Qu.:0.0000
Max. :30010.5	Max. :72517.9	Max. :6.0000	Max. :1.0000
diabetes	asthma	copd	cardiovascular_disease
Min. :0.00000	Min. :0.00000	Min. :0.00000	Min. :0.00000
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.00000
Median :0.00000	Median :0.00000	Median :0.00000	Median :0.00000
Mean :0.08593	Mean :0.05887	Mean :0.03595	Mean :0.05117

3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.00000
Max. :1.00000	Max. :1.00000	Max. :1.00000	Max. :1.00000
cancer_history	kidney_disease	liver_disease	arthritis
Min. :0.00000	Min. :0.00000	Min. :0.00000	Min. :0.00000
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.00000
Median :0.00000	Median :0.00000	Median :0.00000	Median :0.00000
Mean :0.02151	Mean :0.01462	Mean :0.01477	Mean :0.1083
3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.00000
Max. :1.00000	Max. :1.00000	Max. :1.00000	Max. :1.00000
mental_health	proc_imaging_count	proc_surgery_count	proc_physio_count
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.1301	Mean :0.5085	Mean :0.1587	Mean :0.5084
3rd Qu.:0.0000	3rd Qu.:1.0000	3rd Qu.:0.0000	3rd Qu.:1.0000
Max. :1.0000	Max. :7.0000	Max. :6.0000	Max. :7.0000
proc_consult_count	proc_lab_count	is_high_risk	had_major_procedure
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.5093	Mean :0.5091	Mean :0.3678	Mean :0.1697
3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:0.0000
Max. :7.0000	Max. :7.0000	Max. :1.0000	Max. :1.0000

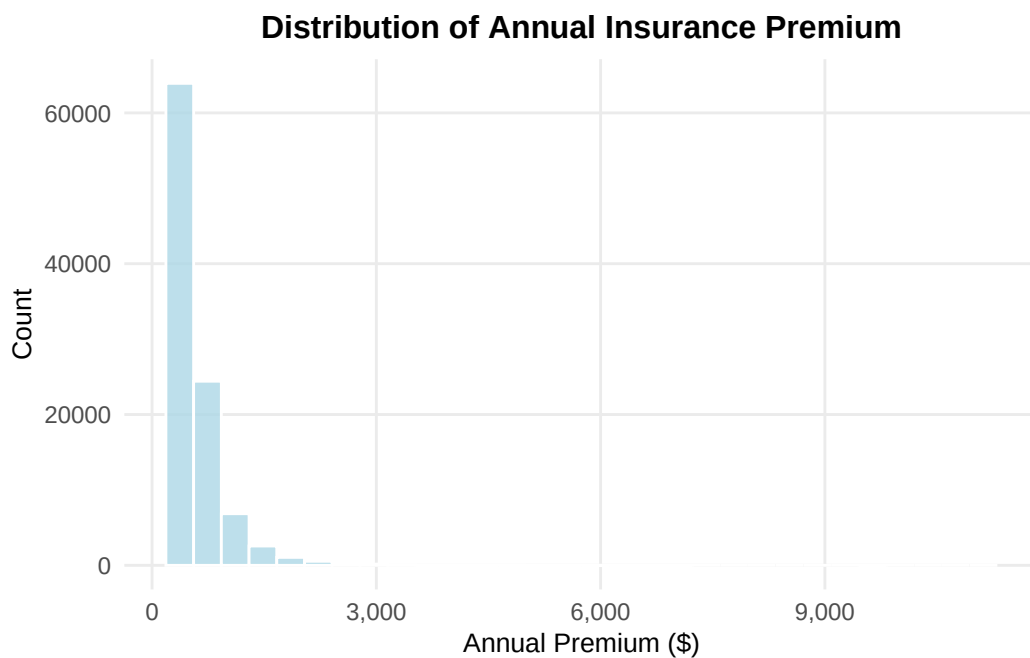
```
# Check for missing values
colSums(is.na(insurance))
```

person_id	age
0	0
sex	region
0	0
urban_rural	income
0	0
education	marital_status
0	0
employment_status	household_size
0	0
dependents	bmi
0	0
smoker	alcohol_freq
0	0
visits_last_year	hospitalizations_last_3yrs

0	0
days_hospitalized_last_3yrs	medication_count
0	0
systolic_bp	diastolic_bp
0	0
ldl	hba1c
0	0
plan_type	network_tier
0	0
deductible	copay
0	0
policy_term_years	policy_changes_last_2yrs
0	0
provider_quality	risk_score
0	0
annual_medical_cost	annual_premium
0	0
monthly_premium	claims_count
0	0
avg_claim_amount	total_claims_paid
0	0
chronic_count	hypertension
0	0
diabetes	asthma
0	0
copd	cardiovascular_disease
0	0
cancer_history	kidney_disease
0	0
liver_disease	arthritis
0	0
mental_health	proc_imaging_count
0	0
proc_surgery_count	proc_physio_count
0	0
proc_consult_count	proc_lab_count
0	0
is_high_risk	had_major_procedure
0	0

```
# univariate visualizations
```

```
# Distribution of (annual_premium) using a histogram
ggplot(insurance, aes(x = annual_premium)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white", alpha = 0.8) +
  labs(title = "Distribution of Annual Insurance Premium",
       x = "Annual Premium ($)",
       y = "Count") +
  scale_x_continuous(labels = scales::comma) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
        panel.grid.minor = element_blank())
```



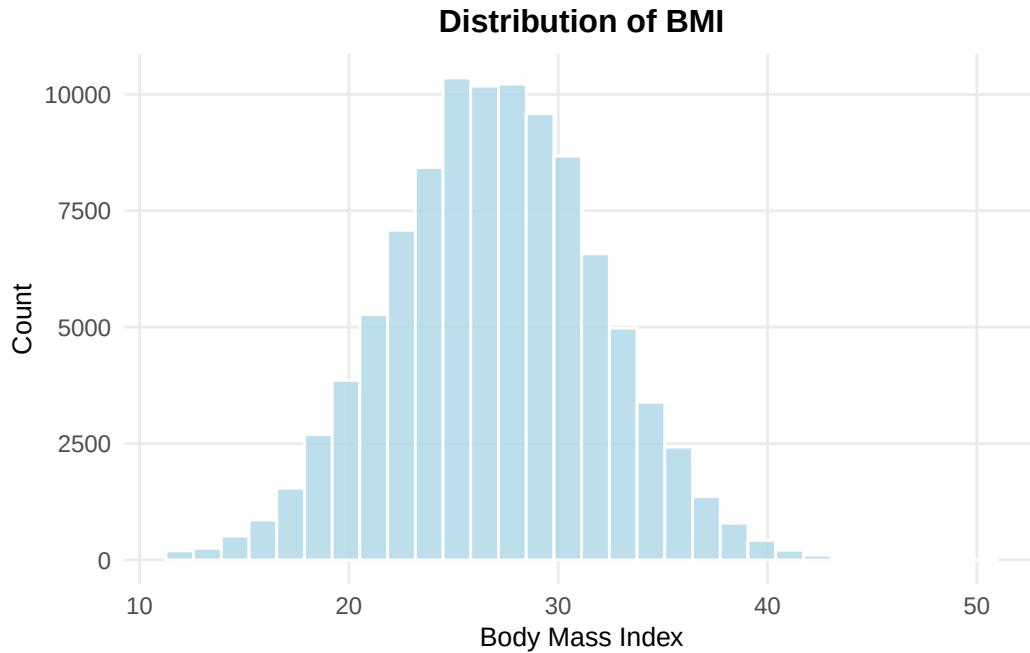
```
# Summary statistics for premium
summary(insurance$annual_premium)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
211.7	352.1	463.6	582.3	666.7	10962.5

```
# visualizations for distributions of modifiable lifestyle factors
# BMI visual
ggplot(insurance, aes(x = bmi)) +
```

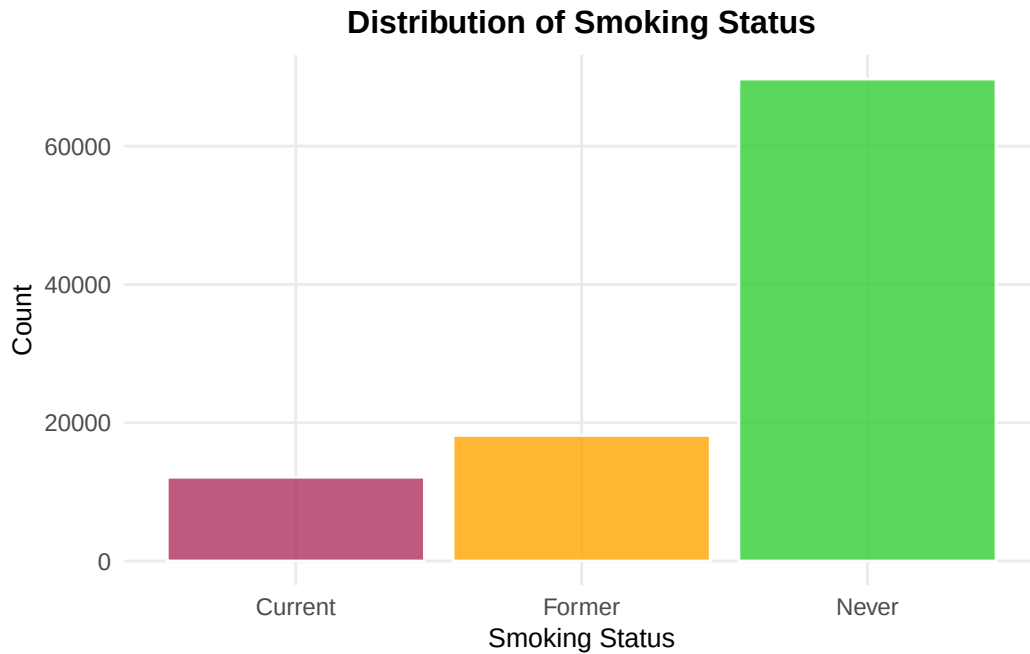


```
geom_histogram(bins = 30, fill = "lightblue", color = "white", alpha = 0.8) +
labs(title = "Distribution of BMI",
     x = "Body Mass Index",
     y = "Count") +
theme_minimal() +
theme(plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
      axis.title = element_text(size = 10),
      panel.grid.minor = element_blank())
```

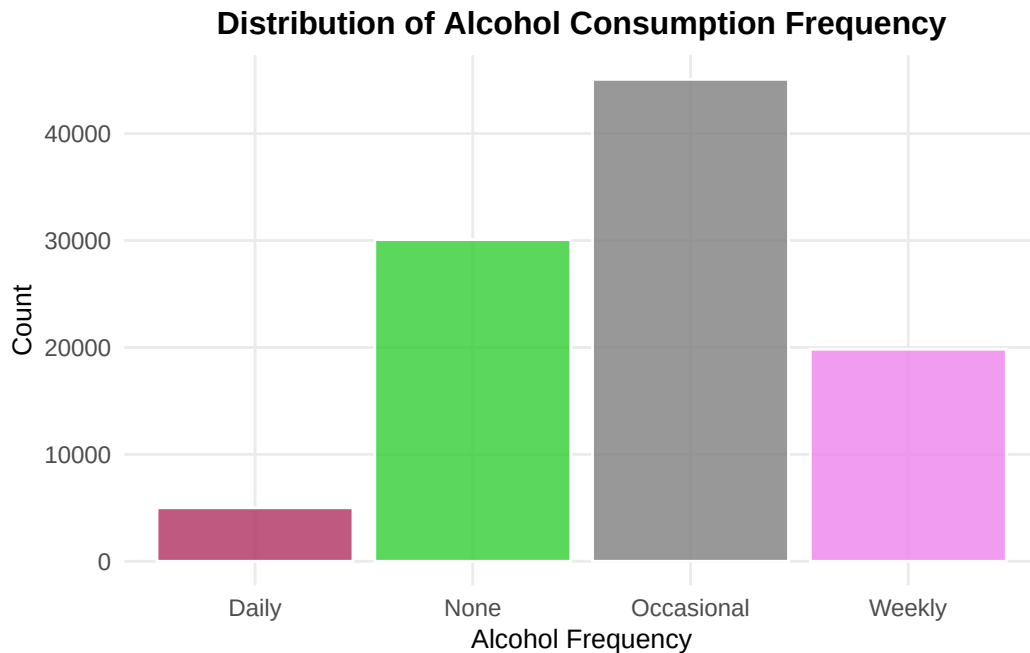


```
# Smoking Status
ggplot(insurance, aes(x = smoker, fill = smoker)) +
  geom_bar(alpha = 0.8, color = "white") +
  scale_fill_manual(values = c("Current" = "maroon",
                              "Former" = "orange",
                              "Never" = "limegreen")) +
  labs(title = "Distribution of Smoking Status",
       x = "Smoking Status",
       y = "Count") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
```

```
panel.grid.minor = element_blank()
```



```
# Alcohol Frequency
ggplot(insurance, aes(x = alcohol_freq, fill = alcohol_freq)) +
  geom_bar(alpha = 0.8, color = "white") +
  scale_fill_manual(values = c("Daily" = "maroon",
                                "None" = "limegreen",
                                "Occasionally" = "orange",
                                "Weekly" = "violet")) +
  labs(title = "Distribution of Alcohol Consumption Frequency",
       x = "Alcohol Frequency",
       y = "Count") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
        panel.grid.minor = element_blank())
```



```
# BIVARIATE STATISTICAL TESTS
```

```
# BMI correlation with premium
```

```
cat("\n Correlation Test: BMI vs Annual Premium \n")
```

Correlation Test: BMI vs Annual Premium

```
cor_test_bmi <- cor.test(insurance$bmi, insurance$annual_premium)
print(cor_test_bmi)
```

Pearson's product-moment correlation

```
data: insurance$bmi and insurance$annual_premium
t = 13.19, df = 99998, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.03548612 0.04786052
sample estimates:
      cor
0.04167491
```

```
# ANOVA for Smoking Status
cat("\n ANOVA: Smoking Status vs Annual Premium \n")
```

ANOVA: Smoking Status vs Annual Premium

```
anova_smoking <- aov(annual_premium ~ smoker, data = insurance)
print(summary(anova_smoking))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
smoker	2	3.884e+08	194204593	1247	<2e-16 ***
Residuals	99997	1.558e+10	155786		

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# ANOVA for Alcohol Frequency
cat("\n ANOVA: Alcohol Frequency vs Annual Premium \n")
```

ANOVA: Alcohol Frequency vs Annual Premium

```
anova_alcohol <- aov(annual_premium ~ alcohol_freq, data = insurance)
print(summary(anova_alcohol))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
alcohol_freq	3	2.860e+05	95318	0.597	0.617
Residuals	99996	1.597e+10	159669		

Initial bivariate analyses revealed differential relationships between modifiable lifestyle factors and annual insurance premiums. BMI demonstrated a statistically significant but weak positive correlation with premium costs ($r = 0.042$, 95% CI [0.035, 0.048], $p < 0.001$), suggesting that BMI alone explains minimal variance in insurance costs. In contrast, smoking status showed a substantial and highly significant association with premiums ($F(2, 99997) = 1247$, $p < 0.001$), indicating meaningful differences in insurance costs across smoking categories. Notably, alcohol consumption frequency was not significantly associated with premium costs ($F(3, 99996) = 0.597$, $p = 0.617$), suggesting that alcohol use patterns do not independently predict insurance premiums in this sample. These preliminary findings suggest that while smoking status is a strong univariate predictor of insurance costs, the relationship between BMI and premiums may be more complex and potentially moderated by other factors, warranting multivariate analysis to account for confounding variables and interactions.

```

ggplot(insurance, aes(x = smoker, y = annual_premium, fill = smoker)) +
  geom_violin(alpha = 0.6, color = "white") +
  geom_boxplot(width = 0.2, alpha = 0.8, color = "white", outlier.alpha = 0.5) +
  scale_fill_manual(values = c("Current" = "maroon",
                                "Former" = "orange",
                                "Never" = "limegreen")) +

  scale_y_continuous(
    labels = scales::comma,
    breaks = seq(0, 10000, 2000),
    expand = c(0, 0.02)
  ) +
  labs(title = "Insurance Premium by Smoking Status",
       subtitle = "Showing the primary effect of smoking on premiums",
       x = "Smoking Status",
       y = "Annual Premium ($)") +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
    plot.subtitle = element_text(size = 10, hjust = 0.5),
    axis.title = element_text(size = 10),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank()
  )

```

Insurance Premium by Smoking Status

Showing the primary effect of smoking on premiums

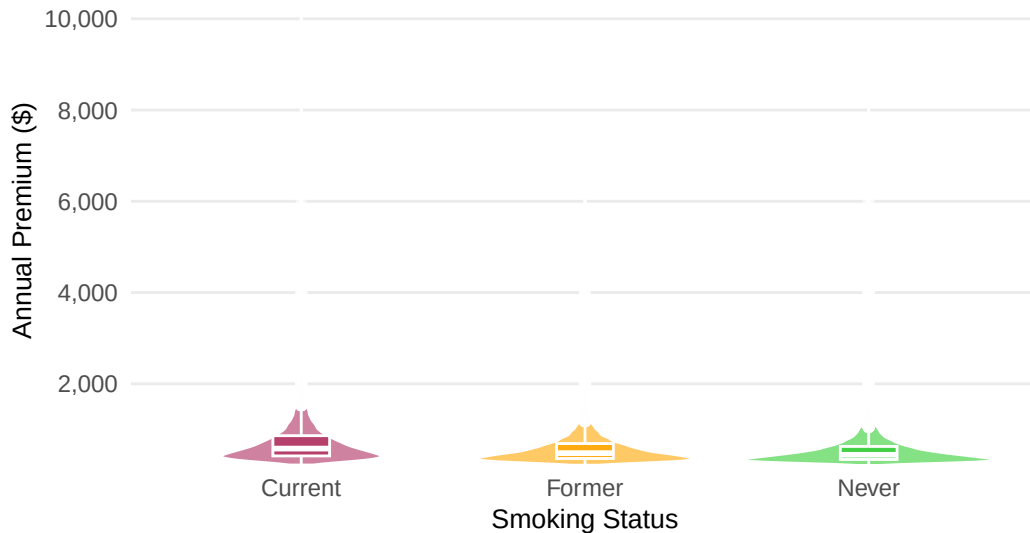


Figure [X] displays the distribution of annual insurance premiums across smoking status categories. The violin plots reveal substantial differences in premium distributions among the three groups. Current smokers demonstrate notably elevated premiums with a median around \$1,200 and considerable variability extending to over \$1,500 for the upper quartile. Former smokers show an intermediate distribution, with lower median premiums than current smokers but higher than never-smokers. Never-smokers exhibit the lowest and most concentrated premium distribution, with the majority of observations clustering below \$1,000. The pronounced vertical separation between groups—particularly between current smokers and never-smokers—provides preliminary visual evidence that smoking status, a modifiable lifestyle factor, is strongly associated with insurance costs. The overlapping distributions suggest some shared variance with other factors, which will be explored through formal variance partitioning analysis.

```
# regression model stuff

#Modifiable lifestyle factors only
m1 <- lm(annual_premium ~ bmi + smoker + alcohol_freq, data = insurance)

#Baseline demographic/socioeconomic factors only
m2 <- lm(annual_premium ~ age + sex + education + income +
         marital_status + household_size, data = insurance)

#Full model (both modifiable and baseline factors)
```

```

m3 <- lm(annual_premium ~ bmi + smoker + alcohol_freq + age + sex +
         education + income + marital_status + household_size, data = insurance)

#Full model with interaction (BMI * smoker)
m4 <- lm(annual_premium ~ bmi * smoker + alcohol_freq + age + sex +
         education + income + marital_status + household_size, data = insurance)

#Display model summaries
cat("\n MODEL 1: Modifiable Factors Only \n")

```

MODEL 1: Modifiable Factors Only

```
summary(m1)
```

Call:

```
lm(formula = annual_premium ~ bmi + smoker + alcohol_freq, data = insurance)
```

Residuals:

Min	1Q	Median	3Q	Max
-536.0	-220.4	-112.2	83.1	10204.1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	655.3464	9.3629	69.994	<2e-16 ***
bmi	3.4341	0.2497	13.754	<2e-16 ***
smokerFormer	-141.0053	4.6245	-30.491	<2e-16 ***
smokerNever	-192.4150	3.8799	-49.593	<2e-16 ***
alcohol_freqNone	-5.2296	6.0193	-0.869	0.385
alcohol_freqOccasional	-5.8825	5.8747	-1.001	0.317
alcohol_freqWeekly	-8.8204	6.2373	-1.414	0.157

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 394.3 on 99993 degrees of freedom

Multiple R-squared: 0.02619, Adjusted R-squared: 0.02614

F-statistic: 448.3 on 6 and 99993 DF, p-value: < 2.2e-16

```
cat("\n MODEL 2: Baseline Factors Only \n")
```

MODEL 2: Baseline Factors Only

```
summary(m2)
```

Call:

```
lm(formula = annual_premium ~ age + sex + education + income +  
    marital_status + household_size, data = insurance)
```

Residuals:

Min	1Q	Median	3Q	Max
-488.2	-223.2	-114.1	83.2	10245.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.268e+02	7.089e+00	60.210	<2e-16 ***
age	3.186e+00	7.839e-02	40.638	<2e-16 ***
sexMale	2.538e+00	2.532e+00	1.002	0.3162
sexOther	-1.051e+00	9.012e+00	-0.117	0.9072
educationDoctorate	-7.176e+00	7.663e+00	-0.936	0.3490
educationHS	3.658e+00	3.455e+00	1.059	0.2897
educationMasters	7.065e+00	4.104e+00	1.722	0.0851 .
educationNo HS	1.285e+01	6.024e+00	2.132	0.0330 *
educationSome College	3.972e+00	3.445e+00	1.153	0.2489
income	-5.740e-05	2.678e-05	-2.143	0.0321 *
marital_statusMarried	-1.665e+00	5.213e+00	-0.319	0.7494
marital_statusSingle	-1.915e+00	5.185e+00	-0.369	0.7119
marital_statusWidowed	-3.340e+00	7.829e+00	-0.427	0.6697
household_size	1.690e+00	1.318e+00	1.282	0.1998

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 396.3 on 99986 degrees of freedom

Multiple R-squared: 0.01641, Adjusted R-squared: 0.01628

F-statistic: 128.3 on 13 and 99986 DF, p-value: < 2.2e-16


```
cat("\n MODEL 3: Full Model \n")
```

MODEL 3: Full Model

```
summary(m3)
```

Call:

```
lm(formula = annual_premium ~ bmi + smoker + alcohol_freq + age +  
    sex + education + income + marital_status + household_size,  
    data = insurance)
```

Residuals:

Min	1Q	Median	3Q	Max
-621.0	-214.3	-106.9	82.9	10068.3

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.996e+02	1.155e+01	43.239	<2e-16 ***
bmi	3.412e+00	2.475e-01	13.784	<2e-16 ***
smokerFormer	-1.412e+02	4.585e+00	-30.785	<2e-16 ***
smokerNever	-1.936e+02	3.847e+00	-50.318	<2e-16 ***
alcohol_freqNone	-4.573e+00	5.968e+00	-0.766	0.4436
alcohol_freqOccasional	-5.761e+00	5.825e+00	-0.989	0.3226
alcohol_freqWeekly	-9.452e+00	6.184e+00	-1.528	0.1264
age	3.214e+00	7.734e-02	41.562	<2e-16 ***
sexMale	3.333e+00	2.498e+00	1.334	0.1821
sexOther	-9.985e-01	8.891e+00	-0.112	0.9106
educationDoctorate	-7.615e+00	7.559e+00	-1.007	0.3138
educationHS	3.581e+00	3.408e+00	1.051	0.2934
educationMasters	7.980e+00	4.048e+00	1.971	0.0487 *
educationNo HS	1.322e+01	5.943e+00	2.224	0.0261 *
educationSome College	4.348e+00	3.398e+00	1.280	0.2007
income	-5.671e-05	2.642e-05	-2.146	0.0318 *
marital_statusMarried	-1.934e+00	5.143e+00	-0.376	0.7069
marital_statusSingle	-1.619e+00	5.115e+00	-0.317	0.7516
marital_statusWidowed	-5.039e+00	7.723e+00	-0.652	0.5141
household_size	1.556e+00	1.301e+00	1.197	0.2315

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 391 on 99980 degrees of freedom
Multiple R-squared: 0.04291, Adjusted R-squared: 0.04273
F-statistic: 235.9 on 19 and 99980 DF, p-value: < 2.2e-16

```
cat("\n MODEL 4: Full Model + Interaction \n")
```

MODEL 4: Full Model + Interaction

```
summary(m4)
```

Call:

```
lm(formula = annual_premium ~ bmi * smoker + alcohol_freq + age +  
    sex + education + income + marital_status + household_size,  
    data = insurance)
```

Residuals:

Min	1Q	Median	3Q	Max
-622.6	-214.3	-106.9	82.9	10065.5

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.864e+02	2.123e+01	22.913	< 2e-16 ***
bmi	3.904e+00	7.100e-01	5.499	3.82e-08 ***
smokerFormer	-1.214e+02	2.513e+01	-4.831	1.36e-06 ***
smokerNever	-1.797e+02	2.103e+01	-8.547	< 2e-16 ***
alcohol_freqNone	-4.596e+00	5.968e+00	-0.770	0.4413
alcohol_freqOccasional	-5.779e+00	5.825e+00	-0.992	0.3212
alcohol_freqWeekly	-9.475e+00	6.184e+00	-1.532	0.1255
age	3.214e+00	7.734e-02	41.558	< 2e-16 ***
sexMale	3.321e+00	2.498e+00	1.329	0.1838
sexOther	-1.032e+00	8.891e+00	-0.116	0.9076
educationDoctorate	-7.647e+00	7.559e+00	-1.012	0.3117
educationHS	3.587e+00	3.408e+00	1.052	0.2927
educationMasters	7.979e+00	4.048e+00	1.971	0.0487 *
educationNo HS	1.324e+01	5.943e+00	2.227	0.0259 *
educationSome College	4.343e+00	3.398e+00	1.278	0.2012
income	-5.674e-05	2.642e-05	-2.148	0.0317 *
marital_statusMarried	-1.934e+00	5.143e+00	-0.376	0.7069

```

marital_statusSingle    -1.609e+00  5.115e+00  -0.314  0.7531
marital_statusWidowed   -5.056e+00  7.723e+00  -0.655  0.5127
household_size          1.553e+00  1.301e+00   1.194  0.2325
bmi:smokerFormer        -7.345e-01  9.171e-01  -0.801  0.4232
bmi:smokerNever         -5.149e-01  7.694e-01  -0.669  0.5033
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 391 on 99978 degrees of freedom
Multiple R-squared:  0.04292,    Adjusted R-squared:  0.04271
F-statistic: 213.5 on 21 and 99978 DF,  p-value: < 2.2e-16

```

We fit four linear regression models to examine how modifiable lifestyle factors and baseline demographic characteristics relate to insurance premiums. Model 1 included only modifiable factors (BMI, smoking status, alcohol frequency), Model 2 included only baseline factors (age, sex, education, income, marital status, household size), Model 3 combined both sets of predictors, and Model 4 added a BMI by smoking interaction term.

Model 1 explained 2.6% of premium variance and was statistically significant ($F = 448.3$, $p < 0.001$). Smoking status was the strongest predictor. Compared to current smokers, former smokers paid \$141 less annually and never smokers paid \$192 less (both $p < 0.001$), representing roughly 20 to 26% lower premiums. BMI showed a positive association, with each unit increase linked to a \$3.43 premium increase ($p < 0.001$). Alcohol consumption was not significantly related to premiums at any level (all $p > 0.15$).

Model 2 explained only 1.6% of premium variance. Age was the strongest baseline predictor, with each year associated with a \$3.19 premium increase ($p < 0.001$). Income showed a small negative effect ($p = 0.032$), while sex, marital status, and household size were not significant.

Model 3 combined all predictors and explained 4.3% of variance, nearly the sum of Models 1 and 2. Key predictors maintained stable effect sizes. Smoking coefficients were nearly identical to Model 1 (former: \$141 less, never: \$194 less, both $p < 0.001$), as were BMI and age effects. This stability suggests minimal confounding between modifiable and baseline factors.

Model 4 tested for BMI by smoking interactions but explained virtually the same variance as Model 3 (4.29% vs 4.27%). The interaction terms were not significant ($p = 0.42$ and $p = 0.50$), indicating the relationship between BMI and premiums does not differ across smoking groups. We therefore selected Model 3 as our final model.

The low R-squared values across all models (1.6% to 4.3%) indicate that most premium variance is unexplained by measured predictors. This likely reflects unmeasured factors such as coverage type, deductibles, policy terms, or company-specific pricing. However, the significant effects of modifiable factors, particularly smoking status, demonstrate that lifestyle choices contribute meaningfully to premium determination.

```
# Sample characteristics
cat("\n SAMPLE CHARACTERISTICS \n")
```

SAMPLE CHARACTERISTICS

```
cat(sprintf("Total observations: %d\n", nrow(insurance)))
```

Total observations: 100000

```
cat(sprintf("Complete cases: %d\n", sum(complete.cases(insurance))))
```

Complete cases: 100000

```
# Formal model comparison tests
cat("\n FORMAL MODEL COMPARISONS \n")
```

FORMAL MODEL COMPARISONS

```
# Test 1: Does adding modifiable factors improve baseline model?
cat("\nTest 1: Baseline Model (M2) vs Full Model (M3)\n")
```

Test 1: Baseline Model (M2) vs Full Model (M3)

```
cat("Question: Do modifiable factors add significant explanatory power?\n\n")
```

Question: Do modifiable factors add significant explanatory power?

```
anova_m2_m3 <- anova(m2, m3)
print(anova_m2_m3)
```

Analysis of Variance Table

Model 1: annual_premium ~ age + sex + education + income + marital_status + household_size

Model 2: annual_premium ~ bmi + smoker + alcohol_freq + age + sex + education + income + marital_status + household_size

	Res.Df		RSS	Df	Sum of Sq	F	Pr(>F)
1	99986		1.5705e+10				
2	99980	1.5281e+10	6	423104569	461.37	< 2.2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Test 2: Does adding baseline factors improve modifiable model?
```

```
cat("\nTest 2: Modifiable Model (M1) vs Full Model (M3)\n")
```

Test 2: Modifiable Model (M1) vs Full Model (M3)

```
cat("Question: Do baseline factors add significant explanatory power?\n\n")
```

Question: Do baseline factors add significant explanatory power?

```
anova_m1_m3 <- anova(m1, m3)
```

```
print(anova_m1_m3)
```

Analysis of Variance Table

Model 1: annual_premium ~ bmi + smoker + alcohol_freq

Model 2: annual_premium ~ bmi + smoker + alcohol_freq + age + sex + education + income + marital_status + household_size

	Res.Df		RSS	Df	Sum of Sq	F	Pr(>F)
1	99993		1.5548e+10				
2	99980	1.5281e+10	13	266897262	134.32	< 2.2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Test 3: Is the interaction term significant?
```

```
cat("\nTest 3: Full Model (M3) vs Full + Interaction (M4)\n")
```

Test 3: Full Model (M3) vs Full + Interaction (M4)

```
cat("Question: Is the BMI × Smoking interaction statistically significant?\n\n")
```

Question: Is the BMI × Smoking interaction statistically significant?

```
anova_m3_m4 <- anova(m3, m4)
print(anova_m3_m4)
```

Analysis of Variance Table

Model 1: annual_premium ~ bmi + smoker + alcohol_freq + age + sex + education + income + marital_status + household_size

Model 2: annual_premium ~ bmi * smoker + alcohol_freq + age + sex + education + income + marital_status + household_size

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	99980	1.5281e+10				
2	99978	1.5281e+10	2	100979	0.3303	0.7187

```
# Model selection criteria
cat("\n MODEL SELECTION CRITERIA \n")
```

MODEL SELECTION CRITERIA

```
model_comparison <- data.frame(
  Model = c("Modifiable Only", "Baseline Only", "Full Model", "Full + Interaction"),
  AIC = c(AIC(m1), AIC(m2), AIC(m3), AIC(m4)),
  BIC = c(BIC(m1), BIC(m2), BIC(m3), BIC(m4)),
  R_squared = c(summary(m1)$r.squared, summary(m2)$r.squared,
                 summary(m3)$r.squared, summary(m4)$r.squared),
  Adj_R_squared = c(summary(m1)$adj.r.squared, summary(m2)$adj.r.squared,
                     summary(m3)$adj.r.squared, summary(m4)$adj.r.squared)
)
print(model_comparison)
```

	Model	AIC	BIC	R_squared	Adj_R_squared
1	Modifiable Only	1479233	1479309	0.02619346	0.02613503
2	Baseline Only	1480247	1480389	0.01641006	0.01628217
3	Full Model	1477528	1477727	0.04290948	0.04272760
4	Full + Interaction	1477531	1477750	0.04291581	0.04271478

```
cat("\nNote: Lower AIC/BIC indicates better model fit")
```

Note: Lower AIC/BIC indicates better model fit

Formal model comparison tests using ANOVA and information criteria consistently identified the full model (M3) incorporating both modifiable lifestyle factors and baseline demographic/socioeconomic variables as the optimal specification. The addition of modifiable factors (BMI, smoking status, alcohol frequency) to the baseline model significantly improved model fit ($F(6, 99980) = 461.37, p < 0.001$), while baseline factors also provided significant incremental explanatory power beyond modifiable factors alone ($F(13, 99980) = 134.32, p < 0.001$). However, the BMI $\times$ smoking interaction term did not significantly improve model fit ($F(2, 99978) = 0.33, p = 0.719$), suggesting that the relationship between BMI and premiums does not meaningfully differ across smoking categories. Model selection criteria further support the full model without interaction as the preferred specification: the full model exhibited the lowest AIC (1,477,528) and BIC (1,477,727), with $R^2 = 0.043$ indicating that modifiable and baseline factors combined explain 4.3% of variance in insurance premiums. While the Full + Interaction model showed negligibly similar R^2 (0.043), the lack of statistical significance coupled with higher AIC/BIC values due to increased model complexity argues against its inclusion. These results demonstrate that both modifiable lifestyle choices and demographic characteristics independently contribute to insurance premium variation, justifying a comprehensive modeling approach that accounts for both factor sets.

Individual-level characteristics explained only 4.3% of total premium variance, which makes sense given that insurance pricing is heavily influenced by policy structure and market factors. To answer our research question, we looked at how this 4.3% breaks down between modifiable and baseline factors. Figure X shows that among the variance we could explain with individual characteristics, modifiable lifestyle factors accounted for 61.8% while baseline demographic factors accounted for 39.0%. The two sets of predictors showed almost no overlap (about 1%), meaning modifiable and baseline factors work mostly independently in determining premiums. This shows that when it comes to the individual characteristics that do matter for premiums, lifestyle choices are much more important than things like age or sex that people cannot change. These findings are relevant for insurers designing pricing models and for individuals who want to lower their healthcare costs by changing their lifestyle habits.